

Emergent $\mathcal{PT}$ Symmetry and Exceptional Points in a Driven Dirac Impurity

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Periodic driving can generate nonequilibrium impurity states, but the microscopic origin of passive non-Hermitian structures is often obscured. We derive such a structure for an inversion-asymmetric spin-orbit-coupled (SOC) Dirac impurity without adding microscopic gain or loss. Cubic crystalline anisotropy selects remote angular harmonics; their off-shell, slowly driven projection produces a real spin-odd shift, and the energy-dependent Dirac bath converts that shift into a relative spin-dependent decay rate. After subtracting common damping, the co-decaying retarded kernel contains the passive parity-time ($\mathcal{PT}$) core $\Delta_{\text{eff}}\sigma_x + i\Gamma_{\text{PT}}\sigma_z$, while its Kramers–Kronig partner supplies a causal detuning floor. The frozen core has $E_{\pm} = \tilde{\epsilon}_{\xi} \pm \sqrt{\Delta_{\text{eff}}^2 - \Gamma_{\text{PT}}^2}$ and an exceptional point (EP) with a finite Jordan simple-plus-double-pole resolvent. We formulate the complete passive problem through its unitary two-channel scattering matrix; its eigenphases obey an exact one-particle contact-phase condition, and their logarithmic derivative equals the Wigner–Smith scattering density-of-states shift. We then isolate a controlled, equal-velocity branchwise linearization. A canonical non-unitary CPT/metric similarity diagonalizes the constant component matrix away from the EP while leaving the scalar hybridization and finite- U interaction invariant. Eliminating the doubly occupied contact amplitude yields the exact Anderson two-electron matrix, rational in a dressed rapidity and covariant under $GL(2, \mathbb{C})$; hence its Yang–Baxter algebra survives the similarity. At the EP the diagonalizer is singular and the same datum is represented instead by an invertible unipotent boundary twist. This integrability statement applies only to the stated linearized submanifold, not to the complete curved, energy-dependent driven kernel. Slave-boson, fluctuation–dissipation, channel-resolved renormalization-group, and operator-resolved biorthogonal NRG checks show pre-EP concentration of the scattering phase density, bounded weak-flow amplification, and a nearly traceless finite- U transition residue aligned with the nilpotent EP direction. The numerical pole splitting establishes neither a stable universal interaction law nor an enhanced thermodynamic Kondo temperature.

I. INTRODUCTION

Driven quantum systems provide a versatile route to steady states and dynamical phases that have no equilibrium analogs. When coherent drive, dissipation, and strong correlations coexist, the resulting many-body dynamics can generate fixed points that reflect both microscopic symmetry and emergent non-Hermitian structure. Recent work has shown that minimizing a driven bosonic field within a Keldysh mean-field framework can reproduce renormalization-group (RG)-like fixed points [1–3], pointing to a deeper connection between nonequilibrium stability and non-Hermitian criticality [4, 5].

Exceptional points (EPs) are non-Hermitian degeneracies where eigenvalues and eigenvectors coalesce, playing a central role in such settings. They reshape spectral topology and enable non-reciprocal transport, sensing enhancement, and dynamical control. EPs have been linked to the non-Hermitian skin effect [6], higher-order degeneracies [7, 8], and ultrafast projective protocols in both quantum and classical experiments [9, 10]. Their influence extends beyond spectra: non-Hermitian topology underpins phenomena in nonlinear photonics [11, 12], topological acoustics [13], engineered metamaterials [14]; coherent light–matter coupling and engineered dissipa-

tion can stabilize correlated states such as topological Kondo phases [15], and nonlinearities can generate instabilities captured by dynamical mean-field theory [16, 17].

Impurity systems are particularly sensitive to anisotropy and dissipation. Boundary-induced openness can mimic potential scattering [18–20], while nonlinear perturbations modify inelastic processes and impurity coherence. Non-Hermitian phase transitions have recently been observed in condensed-matter experiments [21], and nonlinear optical platforms provide precise control of non-Hermitian topology [22–24]. In correlated impurity models, non-Hermitian features manifest through sign reversals in RG flows and causality violations in both interacting and non-interacting settings [4, 25–27]. Non-Hermitian Kondo physics has been explored via numerical renormalization group, cold-atom platforms, and Floquet-engineered conduction bands [6, 28–31]. Recent work on a non-Hermitian Anderson model with complex hybridization has also established a genuine ground-state Kondo-breakdown transition by self-consistent slave-boson and interacting Bethe-Ansatz methods [32]. The present problem is different: its non-Hermitian structure is a passive retarded kernel derived from a Hermitian driven system, and its RG result is deliberately restricted to a weak-flow pre-EP diagnostic. An exact limiting precedent for an emergent non-Hermitian impurity generator was given by Oguri and Sakano [33]: at infinite bias (or in the corresponding high-temperature limit) the Anderson

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impurity maps to a finite Liouville–Fock Hamiltonian, while physical Green functions require the complete doubled-space initial/final-state construction. That high-energy limit is distinct from the present adiabatic passive kernel, but it reinforces the need to retain complete retarded, advanced, and lesser objects rather than assigning observability to an individual left or right eigenvector residue. Integrability of the Kondo model with time-dependent couplings has been established via quantum Knizhnik–Zamolodchikov equations [34], and a $\mathcal{PT}$ -protected Kondo effect has been observed in CeCo_2P_2 [15]. Benchmark studies of the two-dimensional Hubbard model have established the regime of validity of large- N slave-boson and mean-field treatments of correlated electrons [35], directly motivating the approach employed in Sec. V.

The exact algebraic question is separated from the microscopic derivation. The present article derives the driven retarded kernel and identifies the controlled low-energy submanifold on which its contact problem reduces to a finite- U Anderson model. A companion mathematical work [36] develops that reduced model independently: it proves the finite-volume gauge equivalence, YBE/RLL/RTT construction, unipotent boundary-twist limit, the resolution of the nonzero transverse pseudospin-ladder sector, the resulting many-body Jordan structure, and the Laguerre-scaled contraction of descendant spin roots at $\lambda = \infty$. This division of labor is important. The exact result below is not a claim that curvature, unequal branch velocities, or the full frequency-dependent self-energy of the driven model are Yang–Baxter integrable.

Here we show that a driven impurity embedded in a Dirac-like bath naturally develops a passive non-Hermitian kernel once higher angular harmonics and the energy-dependent Dirac bath are treated together. The off-shell auxiliary harmonics generate a real spin-odd coherent shift. The retarded Dirac bath, whose width depends on energy and spin-orbit structure, converts this shift into a spin-dependent decay rate. After subtracting the common bath damping, the remaining co-decaying kernel contains a dynamically induced passive $\mathcal{PT}$ -symmetric core, together with a Kramers–Kronig-related coherent detuning that bounds the approach to exact coalescence. We demonstrate that the resulting finite-drive near-EP structure is compatible with the bath-controlled thermal closure—despite arising without any explicit non-Hermitian term in the microscopic Hamiltonian. The principal spectral figures are frozen-kernel diagnostics. A separate low-temperature $U \rightarrow \infty$ saddle verifies projected-channel scaling and charge continuity, while no finite- r branch is found at the reference temperature $T_b = 0.1$; no finite-temperature enhancement of the condensate is claimed. *In contrast to earlier studies where non-Hermiticity is imposed phenomenologically or emerges through renormalization-group flow [6, 26, 27], we demonstrate a complementary microscopic route in which a strictly Hermitian, periodically driven Dirac impurity produces a passive non-Hermitian relative kernel after coarse graining and removal of the common decay.*

Dirac baths provide a natural setting because spin–momentum locking, angular-momentum selection rules, and an energy-dependent spectral width make the passive $\mathcal{PT}$ channel symmetry constrained.

The paper is organized as follows. Section II defines the full Hermitian Hamiltonian. Section III performs the bilinear Gaussian coarse graining and establishes the emergent $\mathcal{PT}$ symmetry. Section IV derives the steady-state Hamiltonian and the Hadamard rotation that makes hybridization diagonal. Section V presents the slave-boson mean field and the resulting shifted resonance condition. Sections VI–VII state the spectral and fluctuation–dissipation checks, whose full numerical panels are collected in the Supplemental Material. Section VIII discusses real-time dynamics in the passive co-decaying frame. Section IX derives the Jordan response and full scattering eigenphases, then obtains the exact finite- U contact matrix on the controlled branchwise-linearized submanifold and states its CPT-covariant Yang–Baxter structure. Section X separates the analytic frozen estimate from the supporting weak-coupling flow and from any thermodynamic Kondo-temperature claim. Section XI concludes.

II. MODEL AND HAMILTONIAN

The full Hermitian Hamiltonian is $H = H_{\text{bath}} + H_{\text{imp}} + H_{\text{imp-bath}} + H_O + H_{\text{hyb}}[\phi(t)]$. Each term has a distinct physical role. H_{bath} describes the two-dimensional Dirac conduction band with spin-orbit coupling. H_{imp} is the correlated Anderson impurity. $H_{\text{imp-bath}}$ is the bare hybridization between the impurity and the $m = 0$ bath channel. H_O represents auxiliary angular-momentum modes generated by the trigonal crystal anisotropy. H_{hyb} couples the physical bath to the angular harmonics and is modulated by the periodic drive. We define all symbols at first use below. To prevent notation clashes, we reserve $t_{\text{av}} = (t + t')/2$ and $\tau = t - t'$ for the Wigner center and relative times, respectively, and use T_b only for the reservoir temperature. The chiral bath index is $\eta = \pm$, whereas $\zeta_\sigma = +1$ (-1) for $\sigma = \uparrow$ ($\downarrow$). The SOC strength is λ ; β_0 is the signed frozen drive-control amplitude; and $\tilde{\beta} = |\beta_0||b_c|$ is its nonnegative SBMF-renormalized projection. The radial momentum cutoff k_{max} and the ultraviolet energy cutoff D_{uv} are conceptually distinct; in the dimensionless plots both are set numerically to $\pi/4$ as a convention. We denote the analytic local bare-core EP by $\beta_{0,\text{core}}^{\text{EP}} = \gamma = 0.50$. At the SOC-degenerate $k = 0$ point the relevant bath-dressed double-root condition is identical to this core condition. On the finite- k plotting path, the operational Schur-tracked nearest approach is $\beta_{0,\text{proj}}^{\text{min}} = 0.4975$.

A. Bath

We consider a two-dimensional (2D) host crystal with trigonal (C_3) anisotropy. The conduction electrons form a Dirac bath with spin-orbit coupling (SOC):

$$H_{\text{bath}} = \sum_{k,\sigma} \epsilon_k c_{k\sigma}^\dagger c_{k\sigma} + \lambda \sum_k k (c_{k\uparrow}^\dagger c_{k\downarrow} + \text{h.c.}), \quad (1)$$

where $c_{k\sigma}^\dagger$ creates a bath electron with radial momentum k (in units of inverse lattice constant) and spin $\sigma \in \{\uparrow, \downarrow\}$; $\epsilon_k = \alpha_b k^2$ ($\alpha_b > 0$), with radial cutoff $0 \leq k \leq k_{\text{max}} = \pi/4$; and $\lambda > 0$ is the spin-orbit coupling. The separate energy cutoff D_{uv} enters only logarithmic and scaling formulas below. The chiral rotation $c_{k\pm} = (c_{k\uparrow} \pm c_{k\downarrow})/\sqrt{2}$ diagonalizes H_{bath} exactly, yielding SOC-split chiral dispersions $\epsilon_{k\pm} = \alpha_b k^2 \pm \lambda k$ with $\epsilon_{k+} \neq \epsilon_{k-}$ for all $k \neq 0$. Near the Dirac point the bath density of states is $\rho_{\text{bath}}(\omega) \sim |\omega|/\lambda^2$ [4], which is the origin of the pseudogap structure explored in Secs. V–VI.

B. Impurity

The correlated impurity is described by the Anderson Hamiltonian:

$$H_{\text{imp}} = \sum_\sigma \epsilon_\xi d_\sigma^\dagger d_\sigma + U n_\uparrow n_\downarrow, \quad (2)$$

where $d_\sigma^\dagger$ creates an impurity electron with spin σ , $n_\sigma = d_\sigma^\dagger d_\sigma$ is the occupation, ϵ_ξ is the bare impurity level before the projected bath-induced detuning is included, and $U > 0$ is the on-site Coulomb repulsion.

C. Impurity–Bath Hybridization

The impurity couples directly to the $m = 0$ (s-wave) bath channel via

$$H_{\text{imp-bath}} = \sum_{k,\sigma} \left(W_k c_{k,0,\sigma}^\dagger d_\sigma + \text{h.c.} \right), \quad (3)$$

where W_k is the bare hybridization amplitude. This term defines the physical coupling between the impurity and the conduction bath before the angular-momentum harmonics are introduced. Importantly, W_k is time independent and is not proportional to the frozen drive control β_0 . At the slave-boson saddle the physical spinon–bath vertex is rW_k , with $r = |b_c|$.

In the presence of the driven angular-momentum sector [Eqs. (4)–(5)], the $m = 0$ bath propagator becomes dressed by virtual processes involving the auxiliary modes $m \neq 0$: the auxiliary modes generate a retarded self-energy $\Sigma_\sigma^{\text{bath}}$ on the $m = 0$ bath channel (Sec. III). The impurity self-energy then arises from this dressed bath propagator through Eq. (3): it is the auxiliary dressing of the bath propagator, communicated to

the impurity via the bare coupling W_k , that produces the effective non-Hermitian self-energy on the impurity sector. This two-step mechanism—auxiliary modes dress the bath, bath dresses the impurity—is made explicit in Sec. III and Supplemental Material.

D. Auxiliary Angular-Momentum Modes

The trigonal anisotropy $\beta_3(k)k^3 \cos 3\theta$ of the host crystal (where $\beta_3(k)$ is the momentum-dependent trigonal crystal-field strength, distinct from the hybridization amplitude β_0 introduced below; and θ is the in-plane momentum angle) couples the $m = 0$ (s-wave) bath channel to higher angular-momentum harmonics $m = \pm 3, \pm 6, \dots$ via the C_3 selection rule $\Delta m = \pm 3$. These higher harmonics are represented by auxiliary fermions $O_{k,m,\sigma}$ with angular momentum index $m \neq 0$:

$$H_O = \sum_{k,m \neq 0,\sigma} \epsilon_O^{(m)} O_{k,m,\sigma}^\dagger O_{k,m,\sigma} + \sum_{k,m,\sigma} \beta_3(k)k^3 O_{k,m+3,\sigma}^\dagger O_{k,m,\sigma} + \text{h.c.}, \quad (4)$$

where the mode energies satisfy $\epsilon_O^{(m)} = \epsilon_O^{(-m)}$ by mirror/time-reversal symmetry of the C_{3v} crystal field. The fields $O_{k,m,\sigma}$ should be understood as a Feshbach, or Loewdin-partitioned, representation of remote higher-angular-harmonic crystal-field channels rather than as an additional low-energy bath. Their energies include the local orbital or crystal-field splitting of these nonresonant harmonics relative to the active $m = 0$ Dirac channel. In the low-energy limit $\epsilon_O^{(m)} \gg D_{\text{uv}}$, the auxiliary modes are therefore far off-shell and can be integrated out exactly within the quadratic sector; their main effect is a principal-value self-energy dressing the $m = 0$ bath propagator, which in turn induces a self-energy in the impurity sector through the bare hybridization Eq. (3) (see Sec. III and Supplemental Material). If an auxiliary harmonic approaches the conduction-band window, the same formalism gives an ordinary on-shell bath damping contribution; this spin-even damping is not identified with the passive relative $\mathcal{PT}$ channel.

E. Periodic Drive and Hybridization

The periodic drive couples the physical bath ($m = 0$) to the remote higher angular harmonics. We write the Hermitian bath–auxiliary hybridization as

$$H_{\text{hyb}}[\phi(t)] = \sum_{k,\theta,\sigma,m \neq 0} \left[V_m(k) e^{i3m\theta} e^{i\zeta_\sigma \phi(t)} \times c_{k,\theta,\sigma}^\dagger O_{k,m,\sigma} + \text{h.c.} \right]. \quad (5)$$

Here $\zeta_\uparrow = +1$ and $\zeta_\downarrow = -1$ label the spin-helicity channel selected by the spin-orbit-split Dirac bath, while

$\phi(t) = (\pi/4) \sin(\Omega t)$ is the slow drive phase. In the passive mechanism the frozen control is the instantaneous angular velocity $\dot{\phi}(t_{\text{av}}) = (\pi\Omega/4) \cos(\Omega t_{\text{av}})$; the numerical parameter β_0 used below should be read as the corresponding effective frozen-kernel amplitude, not as the value of the phase itself. Hermiticity fixes the second vertex to carry the complex-conjugate phase. Therefore the normal Gaussian contraction of a number-conserving auxiliary sector contains phase differences, not phase sums. This is important: the off-shell auxiliary sector does not by itself generate anti-Hermitian gain or loss.

Equivalently, the phase can be removed from Eq. (5) by the time-dependent unitary rotation

$$O_{k,m,\sigma} \rightarrow e^{-i\zeta_\sigma \phi(t)} O_{k,m,\sigma}. \quad (6)$$

The vertices then become phase-free, while the auxiliary Hamiltonian acquires the Berry term

$$\epsilon_O^{(m)} \rightarrow \epsilon_O^{(m)} - \zeta_\sigma \dot{\phi}(t). \quad (7)$$

Thus a gapped Hermitian auxiliary sector produces, in the adiabatic low-energy projection, a real spin-odd coherent shift. The passive non-Hermitian component appears only in the second step, when this shifted channel is embedded in the energy-dependent retarded Dirac bath. For the Gaussian projection the relevant adiabatic hierarchy is $\Omega \ll \epsilon_O^{(m)}$; the separate Kondo steady-state interpretation is restricted to the finite-drive window where the extracted screening scale exceeds the drive frequency. Because H_O is *bilinear* in $O_{k,m,\sigma}$, the modes $m \neq 0$ can be integrated out at the Gaussian level without approximation within the quadratic sector; the interaction U enters only through the slave-boson mean field on the impurity level (Sec. IV).

III. COARSE GRAINING AND EMERGENT $\mathcal{PT}$ SYMMETRY

This section isolates the bilinear coarse graining from the later slave-boson mean-field treatment of the local interaction. The raw retarded self-energy generated by the auxiliary angular modes is, in general, a complex retarded function; it is not assumed to be positive. The $\mathcal{PT}$ -active low-energy kernel is obtained by projecting the spin-dependent impurity self-energy onto a spin-conjugate pair z, z^* . This formulation keeps the microscopic origin of the passive relative gain-loss channel explicit while avoiding any assumption that the raw retarded envelope is real, positive, or sign definite.

A. Keldysh Path Integral and Angular-Mode Integration

We work within the Keldysh formalism [1], writing the partition function as a coherent-state path integral on

the closed time contour. The bath and auxiliary angular modes enter quadratically and can be eliminated at the Gaussian level. The local Coulomb interaction on the impurity is not treated in this Gaussian step; it is incorporated later through the slave-boson mean-field approximation in Sec. V.

The Grassmann integration can be written explicitly. For fixed (k, θ, σ) , collect the auxiliary fields into a vector $O = (\dots, O_m, \dots)^T$ and write the quadratic contour action as

$$S_O + S_{cO} = \int_C dt dt' \bar{O}(t) A_O(t, t') O(t') + \int_C dt [\bar{c}(t) V_\sigma(t) O(t) + \bar{O}(t) V_\sigma^\dagger(t) c(t)]. \quad (8)$$

The Gaussian identity for Grassmann variables gives

$$\int \mathcal{D}[\bar{O}, O] e^{i(S_O + S_{cO})} \propto \det A_O e^{iS_{\text{infl}}}, \quad (9)$$

$$S_{\text{infl}} = - \int_C dt dt' \bar{c}(t) \Xi_\sigma(t, t') c(t'), \quad (10)$$

$$\Xi_\sigma(t, t') = V_\sigma(t) g_O(t, t') V_\sigma^\dagger(t'). \quad (11)$$

Here $g_O = A_O^{-1}$ is the contour auxiliary propagator. This is the complete Gaussian step: because the auxiliary sector conserves particle number, there is no anomalous OO contraction. Keeping both vertices of Eq. (5) explicit, the normal auxiliary contraction gives

$$\Sigma_\sigma^{O,R}(t, t') = \sum_{m, m' \neq 0} v_{m\sigma}(t) g_{mm'}^{O,R}(t, t') v_{m'\sigma}^*(t'), \quad (12)$$

$$v_{m\sigma}(t) = V_m e^{i3m\theta} e^{i\zeta_\sigma \phi(t)}. \quad (13)$$

For the diagonal terms, and also for any Hermitian off-shell auxiliary matrix after summing conjugate paths, the phases enter as $e^{i\zeta_\sigma [\phi(t) - \phi(t')]}$. With $T = t_{\text{av}}$ and $\tau = t - t'$, this is $e^{i\zeta_\sigma \tau \dot{\phi}(T)}$ to first adiabatic order, so the Wigner transform shifts the stationary auxiliary self-energy as $\Sigma_0^{O,R}(\omega) \rightarrow \Sigma_0^{O,R}[\omega + \zeta_\sigma \dot{\phi}(T)]$. Its first-order expansion gives the regular Hermitian principal-value shift in Eq. (14); the explicit transform and sign are given in the Supplemental Material. Equivalently, after the unitary rotation in Eq. (6), the same result is obtained from the spin-dependent level shift in Eq. (7).

To leading order in the slow drive and in the off-shell expansion, the auxiliary-dressed $m = 0$ bath therefore has the form

$$\Sigma_\sigma^{\text{bath},R}(\omega, t_{\text{av}}) = \Sigma_0^{\text{bath},R}(\omega) + \zeta_\sigma \Delta_{\text{aux}}(\omega, t_{\text{av}}) + O[(\Omega/\epsilon_O)^2]. \quad (14)$$

Here Δ_{aux} is real in the low-energy window. Microscop-

ically one may write, schematically,

$$\Delta_{\text{aux}}(\omega, t_{\text{av}}) = -\dot{\phi}(t_{\text{av}}) \partial_{\epsilon_O} \sum_{m \neq 0} |V_m|^2 \times \mathcal{P} \frac{1}{\omega - \epsilon_O^{(m)}}, \quad (15)$$

up to form-factor and angular-overlap coefficients. If an auxiliary pole enters the conduction window, the corresponding delta-function part gives ordinary bath damping; it is not identified with the passive $\mathcal{PT}$ channel.

The bare impurity–bath hybridization W_k in Eq. (3) remains time independent. The drive reaches the impurity through the dressed bath propagator. In contour notation the two Gaussian eliminations give

$$G_{c,\sigma}^{-1} = g_c^{-1} - \Xi_\sigma, \quad \Xi_\sigma = V_\sigma \circ g_O \circ V_\sigma^\dagger, \quad (16)$$

$$\Sigma_{d,\sigma} = W^\dagger \circ G_{c,\sigma} \circ W, \quad (17)$$

where $\circ$ denotes contour convolution. Thus the drive-dependent virtual path is $d \rightarrow c \rightarrow O \rightarrow c \rightarrow d$; no direct time dependence of W_k is required. The full Schur-complement derivation is given in the Supplemental Material.

In the frozen off-shell window, write $\Xi_\sigma^R = \Xi_0^R + \zeta_\sigma \delta \Xi^R$ with $\delta \Xi^R \in \mathbb{R}$. Expanding the dressed-bath resolvent gives the general first-order correction

$$\delta \Sigma_d^R = W^\dagger G_{c,0}^R \delta \Xi^R G_{c,0}^R W. \quad (18)$$

Although the auxiliary correction is real, the retarded continuum propagators are complex, so the transmitted impurity correction has both coherent and dissipative parts. Defining

$$\Sigma_{d,0}^R(\omega) = \Lambda(\omega) - i\bar{\Gamma}(\omega), \quad (19)$$

we write

$$\Sigma_{d,\sigma}^R = \Sigma_{d,0}^R + \zeta_\sigma \Delta_{\text{coh}} + i\zeta_\sigma \Gamma_{\text{PT}}. \quad (20)$$

For the nearly rigid scalar shift used in the analytic estimates, $\delta \Xi^R \simeq \Delta_{\text{aux}}$ and the spin-even auxiliary background varies slowly. Then

$$\Sigma_{d,\sigma}^R(\omega, t_{\text{av}}) \simeq \Sigma_{d,0}^R(\omega - \zeta_\sigma \Delta_{\text{aux}}) \simeq \Sigma_{d,0}^R - \zeta_\sigma \Delta_{\text{aux}} \partial_\omega \Sigma_{d,0}^R, \quad (21)$$

which gives

$$\Delta_{\text{coh}} = -\Delta_{\text{aux}} \partial_\omega \Lambda, \quad \Gamma_{\text{PT}} = \Delta_{\text{aux}} \partial_\omega \bar{\Gamma}. \quad (22)$$

Equation (18) is the general first-order result; Eq. (21) is its rigid-shift limit. All functions are evaluated in the active low-energy window. Equations (21)–(22) show why the Dirac bath is essential: for a flat wide band with $\partial_\omega \bar{\Gamma} = 0$, the same auxiliary shift would only produce a coherent spin splitting, not a passive $\mathcal{PT}$ decay imbalance. The SOC split bath also supplies the spin-momentum-locked helicity channels. In the chiral basis,

$$\bar{\Gamma}_\eta(\omega) = \pi \sum_k |W_{k\eta}|^2 \delta(\omega - \varepsilon_{c,\eta}(k)), \quad (23)$$

$$\varepsilon_{c,\eta}(k) = \alpha_b k^2 + \eta \lambda k. \quad (24)$$

so the SOC parameter λ controls the channel density of states, the slope $\partial_\omega \bar{\Gamma}_\eta$, and the form factor entering the projected EP diagnostics. Without SOC-induced helicity structure and energy-dependent Dirac spectral weight, the auxiliary shift remains a coherent real field rather than a passive relative decay channel.

B. Passive $\mathcal{PT}$ Core and Causal Detuning

The projected impurity self-energy is decomposed into a common bath part, a passive relative decay imbalance, and a Kramers–Kronig-related coherent detuning:

$$\Sigma_{\text{imp}}^R(\omega, t_{\text{av}}) = [\Lambda(\omega) - i\bar{\Gamma}(\omega)] \mathbb{I} + \Delta_{\text{coh}}(\omega, t_{\text{av}}) \sigma_z + i\Gamma_{\text{PT}}(\omega, t_{\text{av}}) \sigma_z. \quad (25)$$

The first line is a common Lamb shift and common decay. It is retained in the bath-controlled FDR, but it does not affect the relative EP structure. The second line contains two spin-odd terms with different symmetry status: $i\Gamma_{\text{PT}} \sigma_z$ is the passive relative decay imbalance, while $\Delta_{\text{coh}} \sigma_z$ is a real coherent detuning generated by the same causal retarded bath self-energy.

The SOC bath supplies the Hermitian spin-mixing channel. In the original spin basis, the local bath Green function has the matrix structure

$$g_b^R(k, \omega) = [\omega^+ - \epsilon_k - \lambda k \sigma_x]^{-1} = g_0(k, \omega) \mathbb{I} + g_x(k, \omega) \sigma_x, \quad (26)$$

with

$$g_x(k, \omega) = \frac{\lambda k}{(\omega^+ - \epsilon_k)^2 - \lambda^2 k^2}. \quad (27)$$

The principal-value part of this SOC off-diagonal bath dressing gives a real coherent spin-mixing coefficient,

$$\Delta_{\text{eff}}(\omega) = \sum_k |W_k|^2 \text{Re } g_x(k, \omega), \quad (28)$$

renormalized in the interacting problem by the slave-boson saddle point. The drive-induced shift cannot leak into this mixing channel at leading order: expanding the dressed bath propagator to first order in the spin-odd shift $\Delta_{\text{aux}} \sigma_z$ gives

$$g_b^R \sigma_z g_b^R = (g_0^2 - g_x^2) \sigma_z, \quad (29)$$

because $\{\sigma_x, \sigma_z\} = 0$, so the first-order correction is purely σ_z and Δ_{eff} remains a static (drive-independent) channel at this order, corresponding to the offset Δ_0 of Eq. (36); the isolated finite-drive EP of the generic projection is therefore the microscopically natural case. Thus, after subtracting the common damping, the general relative two-channel kernel is

$$h_{\text{rel}} = \Delta_{\text{eff}} \sigma_x + (\Delta_{\text{coh}} + i\Gamma_{\text{PT}}) \sigma_z. \quad (30)$$

The passive- $\mathcal{PT}$ -symmetric core is obtained at the compensated detuning point $\Delta_{\text{coh}} = 0$:

$$h_{\mathcal{PT}} = \Delta_{\text{eff}}\sigma_x + i\Gamma_{\text{PT}}\sigma_z, \quad \sigma_x h_{\mathcal{PT}}^* \sigma_x = h_{\mathcal{PT}}. \quad (31)$$

The real term $\Delta_{\text{coh}}\sigma_z$ is not discarded. It is $\mathcal{PT}$ -odd under the same operation and therefore acts as a causal detuning floor. The pole splitting of Eq. (30) is

$$s^2 = \Delta_{\text{eff}}^2 + (\Delta_{\text{coh}} + i\Gamma_{\text{PT}})^2. \quad (32)$$

For real Δ_{eff} , Δ_{coh} , and Γ_{PT} , exact coalescence in the passive sector requires

$$\Delta_{\text{coh}}\Gamma_{\text{PT}} = 0, \quad \Delta_{\text{eff}}^2 + \Delta_{\text{coh}}^2 - \Gamma_{\text{PT}}^2 = 0. \quad (33)$$

Thus, for a nonzero relative decay imbalance, the exact passive EP occurs at the compensated point $\Delta_{\text{coh}} = 0$ and $|\Delta_{\text{eff}}| = |\Gamma_{\text{PT}}|$. Compensation is an operating-energy condition rather than a fine tuning: $\Delta_{\text{coh}} \propto \partial_\omega \Lambda$ changes sign and vanishes at a finite energy of order D_{uv}/e for the linear Dirac width, where $\partial_\omega \Gamma$ remains finite, while near the Fermi level the companion ratio is logarithmically enhanced, $|\Delta_{\text{coh}}/\Gamma_{\text{PT}}| \simeq (2/\pi) \ln(D_{\text{uv}}/|\omega|)$. Away from the compensated point, the Kramers–Kronig detuning prevents a literal pole coalescence. Minimizing $|s|$ over real Δ_{eff} at fixed Δ_{coh} and Γ_{PT} gives

$$|s|_{\min} = \begin{cases} \sqrt{2|\Delta_{\text{coh}}\Gamma_{\text{PT}}|}, & |\Delta_{\text{coh}}| \leq |\Gamma_{\text{PT}}|, \\ \sqrt{\Delta_{\text{coh}}^2 + \Gamma_{\text{PT}}^2}, & |\Delta_{\text{coh}}| > |\Gamma_{\text{PT}}|, \end{cases} \quad (34)$$

with the square-root non-analyticity characteristic of a second-order EP. This causal detuning should not be identified with a universal interaction-induced splitting. In the exact equal-velocity, scalar-hybridization linearized reduction derived below, the standard local interaction preserves the complexified pseudospin symmetry and does not lift the EP [36]. Curvature, unequal velocities, an energy-dependent self-energy, or component-dependent vertices can instead move or split it. The Supplemental Material derives the minimization and the sharp-cutoff Dirac Kramers–Kronig estimate explicitly. The figures should therefore be read as near-EP diagnostics of the co-decaying projected kernel, with the causal scale $s_{\min}$ preventing an unphysical pole on the real-frequency axis.

This two-step structure is the microscopic origin of the kernel used in the rest of the paper: off-shell auxiliary modes generate a real spin-odd shift; the energy-dependent Dirac bath converts that shift into a spin-dependent width; and the SOC bath provides the Hermitian spin-mixing channel. The numerical scans treat β_0 as a tunable frozen effective kernel control parameter. Microscopically,

$$\Gamma_{\text{PT}} \sim \Omega \frac{|V|^2}{\epsilon_0^2} \partial_\omega \bar{\Gamma}, \quad (35)$$

so the construction establishes the symmetry-allowed origin of the passive relative channel, but does not determine the numerical magnitude of the dimensionless

frozen-kernel control β_0 . Figures 1 and 2 therefore use a normalized geometry scan to expose the coalescence structure. The independent causal test in Supplemental Fig. S6 instead uses a fixed small bare width $\Gamma_0 = 0.12$ relative to the unit charge gap. This distinction prevents the adiabatic estimate in Eq. (35) from being misread as a microscopic prediction of $\beta_0 \simeq 0.5$.

IV. STEADY-STATE HAMILTONIAN AND EMERGENT NON-HERMITICITY

Given the passive spin-conjugate relative self-energy from Sec. III, this section derives the explicit 4×4 steady-state Hamiltonian $\tilde{H}(k)$ and identifies the structural origin of its exceptional points. A key step is the Hadamard rotation that diagonalizes the bare hybridization block, revealing that each chiral spinon mode couples directly to one bath channel. The internal $i\Gamma_{\text{PT}}$ term still mixes those impurity modes, so the physical two-channel scattering matrix need not be diagonal; the frozen finite-size quantization below is therefore formulated using its exact eigenphases.

A. From Self-Energy to Steady-State Hamiltonian

After the Gaussian coarse-graining of the bath and auxiliary angular modes in Sec. III, and after introducing the slave-boson saddle point in Sec. V, the retarded spinon action takes the form $S_{\text{imp}}^R = \sum_k \bar{f}[\omega\mathbb{I} - H_{\text{ss}}(k)]f$, where the steady-state Hamiltonian $H_{\text{ss}}(k)$ is defined as the inverse of the retarded Green's function at the saddle point: $H_{\text{ss}}(k) \equiv [G_{\text{ss}}^R(\omega, k)]^{-1}|_{\text{saddle}}$. The gauge-invariant slave-boson amplitude is $r = |b_c|$. The three channels in the effective Hamiltonian have distinct microscopic origins: the passive relative width and coherent detuning are components of the projected retarded self-energy, whereas the impurity–bath vertex descends from the time-independent bare coupling W . Since an impurity self-energy contains two spinon–bath vertices, the physical projected parametrization is

$$\Gamma_{\text{PT}} = r^2 \Gamma_{\text{PT}}^{(0)}(\beta_0), \quad \Delta_{\text{eff}} = r^2 [\Delta_0^{(0)} + c_\Delta^{(0)} \beta_0], \quad V_{\text{eff}} = rW. \quad (36)$$

Here the superscript (0) denotes the frozen bath projection before slave-boson renormalization. The core discriminant consequently obeys

$$s_{\text{core}}^2 = r^4 \left\{ [\Delta_0^{(0)} + c_\Delta^{(0)} \beta_0]^2 - [\Gamma_{\text{PT}}^{(0)}(\beta_0)]^2 \right\}. \quad (37)$$

For $r > 0$, the common factor r^4 cancels from the core EP condition. The local core EP is therefore exactly invariant under changes of r , whereas a bath-dressed finite- k Schur minimum must still be tracked. Figures 1 and 2 use $r = 1$ and a normalized frozen-kernel geometry path;

they are intended to display the bifurcation rather than to fix an Anderson hybridization width. The supporting channel-resolved RG calculation uses the conservative fixed- W projection

$$\Gamma_0 = \pi\rho W^2 = 0.12, \quad \Delta_{\text{eff}}^{(0)} = 0.075 + 0.05\beta_0, \quad \Gamma_{\text{PT}}^{(0)} = 0.20\beta_0, \quad (38)$$

for which $\Gamma_0/\Delta_{\text{charge}} = 0.12$ and the core EP remains at $\beta_0 = 0.50$. Thus the RG test is performed in a local-moment-compatible small-width regime, with W independent of the drive coordinate.

At the compensated passive- $\mathcal{PT}$ point $\Delta_{\text{coh}} = 0$, substituting the projected self-energy into the retarded Dyson equation and including the hybridization to the chiral bath channels gives, in the spin basis $(d_\uparrow, d_\downarrow, c_{k+}, c_{k-})$,

$$H_{\text{ss}}(k) = \begin{pmatrix} \tilde{\epsilon}_\xi + i\Gamma_{\text{PT}} & \Delta_{\text{eff}} & V_{\text{eff}}/\sqrt{2} & V_{\text{eff}}/\sqrt{2} \\ \Delta_{\text{eff}} & \tilde{\epsilon}_\xi - i\Gamma_{\text{PT}} & V_{\text{eff}}/\sqrt{2} & -V_{\text{eff}}/\sqrt{2} \\ V_{\text{eff}}/\sqrt{2} & V_{\text{eff}}/\sqrt{2} & \varepsilon_{c,+}(k) & 0 \\ V_{\text{eff}}/\sqrt{2} & -V_{\text{eff}}/\sqrt{2} & 0 & \varepsilon_{c,-}(k) \end{pmatrix}. \quad (39)$$

Here $\tilde{\epsilon}_\xi = \epsilon_\xi + \delta_c + \delta_\Sigma$ is the renormalized impurity level. The Lagrange multiplier δ_c enforces the slave-boson constraint, while δ_Σ is the real projected one-body shift from the principal-value self-energy in Sec. III B (and may be absorbed into ϵ_ξ by convention). The chiral bath dispersions are $\varepsilon_{c,\pm}(k) = \alpha_b k^2 \pm \lambda k$. The interaction U is not an explicit two-body coupling in the 4×4 Gaussian kernel; it enters through the SBMF constraint and through the Schrieffer-Wolff and finite- U contact virtual-charge energies used below. For the finite- U Anderson interpretation, particle-hole symmetry is imposed by the common impurity-level choice $\epsilon_\xi = -U/2$, not by adding opposite $\pm U/2$ spin splittings. A real Zeeman field, when present, would enter as an independent spin splitting B_{imp} rather than as the Hubbard interaction.

The $\mathcal{PT}$ symmetry of $H_{\text{ss}}(k)$ is explicit for real Γ_{PT} , Δ_{eff} , and V_{eff} : under $P : \uparrow \leftrightarrow \downarrow$ combined with complex conjugation, $(\tilde{\epsilon}_\xi + i\Gamma_{\text{PT}}) \rightarrow (\tilde{\epsilon}_\xi - i\Gamma_{\text{PT}})^* = \tilde{\epsilon}_\xi + i\Gamma_{\text{PT}}$, while the real off-diagonal and hybridization entries are exchanged into themselves. Thus the $\mathcal{PT}$ structure is fixed by the projected self-energy channels and the gauge-invariant amplitude $r = |b_c|$, not by the phase of the slave-boson condensate.

B. Diagonal-Hybridization Rotation

The hybridization block of H_{ss} has Hadamard structure $(V_{\text{eff}}/\sqrt{2})\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. The unitary rotation $U_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ acting on the impurity subspace defines chiral spinon modes $\tilde{d}_\pm = (d_\uparrow \pm d_\downarrow)/\sqrt{2}$ and diagonalizes

the hybridization block exactly:

$$U_2^{-1} h_{\text{imp}} U_2 = \begin{pmatrix} \tilde{\epsilon}_\xi + \Delta_{\text{eff}} & i\Gamma_{\text{PT}} \\ i\Gamma_{\text{PT}} & \tilde{\epsilon}_\xi - \Delta_{\text{eff}} \end{pmatrix}, \quad (40)$$

$$U_2^{-1} V_{dc} = V_{\text{eff}} \mathbb{I}, \quad (41)$$

where $h_{\text{imp}} = \begin{pmatrix} \tilde{\epsilon}_\xi + i\Gamma_{\text{PT}} & \Delta_{\text{eff}} \\ \Delta_{\text{eff}} & \tilde{\epsilon}_\xi - i\Gamma_{\text{PT}} \end{pmatrix}$ and $V_{dc} = (V_{\text{eff}}/\sqrt{2})\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. The algebra is identical to the symmetric case: $\frac{V_{\text{eff}}}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^2 = \frac{V_{\text{eff}}}{2} \cdot 2\mathbb{I} = V_{\text{eff}}\mathbb{I}$. The full rotated Hamiltonian in the $(\tilde{d}_+, \tilde{d}_-, c_{k+}, c_{k-})$ basis is:

$$\tilde{H}(k) = \begin{pmatrix} \tilde{\epsilon}_\xi + \Delta_{\text{eff}} & i\Gamma_{\text{PT}} & V_{\text{eff}} & 0 \\ i\Gamma_{\text{PT}} & \tilde{\epsilon}_\xi - \Delta_{\text{eff}} & 0 & V_{\text{eff}} \\ V_{\text{eff}} & 0 & \varepsilon_{c,+}(k) & 0 \\ 0 & V_{\text{eff}} & 0 & \varepsilon_{c,-}(k) \end{pmatrix}. \quad (42)$$

Three key properties of $\tilde{H}(k)$ follow:

- (i) *Diagonal hybridization*: $\tilde{d}_+$ hybridizes exclusively with the c_{k+} bath (amplitude V_{eff}) and $\tilde{d}_-$ with c_{k-} , independently. There is no bare cross-channel vertex. The internal $i\Gamma_{\text{PT}}$ matrix element can nevertheless mediate conversion between the two asymptotic bath channels, so it does not imply a diagonal physical scattering matrix.
- (ii) *Impurity block*: the 2×2 frozen impurity block $h_d = \begin{bmatrix} \tilde{\epsilon}_\xi + \Delta_{\text{eff}} & i\Gamma_{\text{PT}} \\ i\Gamma_{\text{PT}} & \tilde{\epsilon}_\xi - \Delta_{\text{eff}} \end{bmatrix}$ contains a real coherent splitting Δ_{eff} and an anti-Hermitian relative gain-loss channel $i\Gamma_{\text{PT}}$. Both are measured in units of the gauge-invariant scale $\tilde{\beta} = |\beta_0||b_c|$, but their projection coefficients need not be identical.
- (iii) *EP structure*: h_d has eigenvalues $E_\pm = \tilde{\epsilon}_\xi \pm s$ with $s = \sqrt{\Delta_{\text{eff}}^2 - \Gamma_{\text{PT}}^2}$. In the frozen diagnostic notation of Sec. IX, this is written as $s = \sqrt{\gamma^2 - \beta_0^2}$, with $\gamma \leftrightarrow \Delta_{\text{eff}}$ and $\beta_0 \leftrightarrow \Gamma_{\text{PT}}$ treated as independent coherent and relative gain-loss amplitudes. The local EP condition is $|\Gamma_{\text{PT}}| = |\Delta_{\text{eff}}|$; V_{eff} controls bath hybridization and bath-induced unfolding of this local condition.

a. Full 4×4 eigenvalue condition. The numerical eigenvalues and condition numbers in Figs. 1 and 2 are computed from the full channel-projected kernel of Eq. (42), not from the isolated impurity block alone. Eliminating the bath components gives the exact Schur complement condition

$$\mathcal{P}(z, k) = \mathbf{p}_+(z, k)\mathbf{p}_-(z, k) + \Gamma_{\text{PT}}^2 = 0, \quad (43)$$

where

$$\mathbf{p}_\eta(z, k) = z - \tilde{\epsilon}_\xi - \eta\Delta_{\text{eff}} - \frac{V_{\text{eff}}^2}{z - \varepsilon_{c,\eta}(k)}, \quad \eta = \pm. \quad (44)$$

Equivalently, after clearing denominators, the full eigenvalues are roots of the quartic equation

$$[\delta_{d,+}(z, k)\delta_{c,+}(z, k) - V_{\text{eff}}^2][\delta_{d,-}(z, k)\delta_{c,-}(z, k) - V_{\text{eff}}^2] + \Gamma_{\text{PT}}^2\delta_{c,+}(z, k)\delta_{c,-}(z, k) = 0, \quad (45)$$

where

$$\delta_{d,\eta}(z, k) = z - \tilde{\epsilon}_\xi - \eta\Delta_{\text{eff}}, \quad \delta_{c,\eta}(z, k) = z - \epsilon_{c,\eta}(k). \quad (46)$$

Equation (45) is only the characteristic polynomial of the quadratic 4×4 kernel; it is unrelated to an interaction-induced quartic scaling law for the EP pole splitting. The full EP condition is therefore

$$\mathcal{P}(z_{\text{EP}}, k) = 0, \quad \partial_z \mathcal{P}(z_{\text{EP}}, k) = 0. \quad (47)$$

The simpler impurity-block expression in item (iii) is the transparent $V_{\text{eff}} \rightarrow 0$ limit and explains the local square-root mechanism; the plotted spectra use the full 4×4 matrix. At $k = 0$, however, the SOC-split bath energies coincide and the Schur self-energy is a common scalar shift $\zeta(z) = z - \tilde{\epsilon}_\xi - V_{\text{eff}}^2/z$. Equation (43) then reduces to

$$\mathcal{P}(z, 0) = \zeta(z)^2 - (\Delta_{\text{eff}}^2 - \Gamma_{\text{PT}}^2). \quad (48)$$

The impurity-related double root therefore occurs exactly at $\Delta_{\text{eff}}^2 = \Gamma_{\text{PT}}^2$, independent of W and of $r > 0$. SOC-resolved finite- k asymmetry breaks this common-shift form, so its operational minimum can drift and must be tracked rather than called an exact core EP. An operator-resolved finite- U biorthogonal NRG calculation for the corresponding scalar-hybridization Wilson-chain control is consistent with this hierarchy: the transition-residue matrix remains aligned with the nilpotent EP direction and is nearly traceless, whereas the branch-resolved pole splitting does not yield a stable universal interaction exponent or quartic coefficient (Supplemental Fig. S8). In the opposite limit in which one chiral impurity-bath channel is isolated, the square root is

$$E_{\eta,\pm} = \frac{\epsilon_d + \epsilon_{k\eta}}{2} \pm \frac{1}{2}\sqrt{(\epsilon_d - \epsilon_{k\eta})^2 + 4V_\eta^R V_\eta^L}. \quad (49)$$

For a positive Hermitian product $V_\eta^R V_\eta^L > 0$ this gives the usual avoided crossing. If the product is negative or complex, the discriminant can instead vanish, producing a hybridization-induced non-Hermitian EP or crossover. In the revised gauge-fixed formulation, however, such a complex or sign-changing hybridization product is not identified with a Kondo scale unless it contributes to a positive screening eigenvalue of the biorthogonal exchange matrix.

b. Behaviour at $\beta_0 = 0$. In the minimal symmetric representative, the exact origin has $\Gamma_{\text{PT}} = \Delta_{\text{eff}} = 0$ while the time-independent hybridization rW remains common to both channels. At the SOC-degenerate $k = 0$ point that common bath dressing is proportional to the identity, so the local impurity block reduces to $\tilde{\epsilon}_\xi \mathbb{I}$. This is

a normal twofold degeneracy, not a genuine exceptional point: the matrix is diagonalizable and one may choose an orthonormal eigenbasis with $\kappa_{\text{imp}} = 1$. The apparent singularity of an EP-adapted normalization near this point is therefore a path-dependent diagnostic effect, not a physical finite-drive EP. In a generic projection with $\Delta_0 \neq 0$, the origin is instead a normal split impurity doublet, and an isolated local impurity EP occurs only when $|c_\Gamma \tilde{\beta}| = |\Delta_0 + c_\Delta \tilde{\beta}|$. Thus a finite offset or nonlinear channel projection converts the fully symmetric EP ray into an isolated finite-drive EP. Along a regular off-EP path the basis returns smoothly to a normal one and $\kappa_{\text{imp}} \rightarrow 1$.

The local interaction U does not by itself lift this trivial origin degeneracy in the spin-symmetric saddle point: it mainly renormalizes the scalar impurity level $\tilde{\epsilon}_\xi$ and the condensate amplitude. Only a spin-asymmetric term, such as an external Zeeman field or a spin-polarized real one-body shift proportional to $\langle n_\downarrow \rangle - \langle n_\uparrow \rangle$, would split the origin already at $\beta_0 = 0$. For this reason, the finite- $|\beta_0|$ condition-number peaks, where eigenvalue bifurcation and eigenvector coalescence occur away from the trivial normal degeneracy, are the physically relevant EP diagnostics in Figs. 1 and 2.

c. Interaction and the EP. The local Hubbard interaction is a scalar in the two-component Fock space, $n_{d+}n_{d-} = N_d(N_d - 1)/2$. It therefore does not by itself break the complexified pseudospin symmetry of the equal-velocity, scalar-hybridization linearized model. On that exact submanifold the EP remains symmetry protected at finite U and $\kappa_{\text{imp}} \sim C/|s|$. In the complete driven problem, deviations from that submanifold may shift or split the EP; no universal interaction “floor” is inserted here.

V. SLAVE-BOSON MEAN FIELD AND KELDYSH SELF-CONSISTENCY

The effective impurity problem originates from a correlated Anderson level with local repulsion U . We formulate the constrained low-energy sector using the $U \rightarrow \infty$ large- N slave-boson mean-field saddle [31, 37, 38]. The finite- $U = 2$ Schrieffer-Wolff formulas used later are a separate frozen analytic estimate. This section introduces the spinon operators $f_\sigma^\dagger$ that replace the physical $d_\sigma^\dagger$ after the slave-boson decoupling, derives the self-consistent equations for the condensate amplitude $r = |b_c|$ and Lagrange multiplier δ_c , and formulates the shifted resonance condition in gauge-invariant language.

A. Slave-Boson Representation

To handle the Coulomb repulsion U in the Keldysh framework, we use the large- N slave-boson (SBMF) rep-

resentation [31, 37, 38]:

$$d_\sigma^\dagger = f_\sigma^\dagger b, \quad b^\dagger b + \sum_\sigma f_\sigma^\dagger f_\sigma = Q. \quad (50)$$

Here $f_\sigma^\dagger$ is the auxiliary spinon fermion carrying spin σ (the physical quasiparticle in the Kondo limit), and $b^\dagger$ is the slave boson enforcing no double occupancy. The parameter Q fixes the constraint normalization: $Q = 1$ gives the physical spin-1/2 infinite- U constraint, while $Q = Nq_{\text{occ}}$ denotes the large- N or channel-rescaled normalization. All saddle-point amplitudes quoted below are interpreted in the same normalization. The constraint is imposed via a Lagrange multiplier δ_c . At the saddle point $b \rightarrow b_c \in \mathbb{C}$, the phase of b_c is an internal $U(1)$ gauge choice. We therefore work in the gauge-fixed convention in which observables depend on the amplitude $r \equiv |b_c|$, and the spinon hybridization scale is rW_k .

The renormalized impurity level and gauge-invariant SBMF scale are then

$$\tilde{\epsilon}_\xi = \epsilon_\xi + \delta_c + \delta_\Sigma, \quad (51)$$

$$\tilde{\beta} = |\beta_0| |b_c| = |\beta_0| r \quad (52)$$

Here δ_Σ is the real principal-value one-body shift of the projected impurity level; if this shift is absorbed into the bare level, one may set $\delta_\Sigma = 0$ and reinterpret ϵ_ξ accordingly. The projected channels entering H_{ss} are defined by Eq. (36); in particular, the hybridization is $V_{\text{eff}} = rW$ rather than $c_V\beta$. The equations below define the large- N saddle. For the fixed- W small-width path of Eq. (38), a seed-independent finite- r solution is found at $T_b = 10^{-6}$ throughout $0.35 \leq \beta_0 \leq 0.55$: r decreases smoothly from 0.01047 to 0.00955, with maximum constraint and stationarity residuals below 4.1×10^{-10} and 9.6×10^{-10} , respectively. Charge continuity is satisfied to 3.9×10^{-23} . At the manuscript reference temperature $T_b = 0.1$, however, the same $U \rightarrow \infty$ closure has no finite- r branch and terminates at the $r = 0$ boundary. The spectra displayed below therefore use stated frozen values and are not used to claim a finite-temperature enhancement of r . The complete convergence and seed-independence audit is given in the Supplemental Material.

B. Saddle-Point Equations and Gauge-Invariant Amplitude

The Keldysh saddle-point conditions follow by varying the complete contour action with respect to $r = |b_c|$ and δ_c (Supplemental Material):

$$0 = r^2 + n_f - Q, \quad n_f = -i \int \frac{d\omega}{2\pi} \text{Tr}_s G_{ff}^<, \quad (53)$$

$$0 = 2r\delta_c + \mathcal{F}_W + 2rD_0\mathcal{X} + 2r\mathcal{I}_\Gamma. \quad (54)$$

The hybridization force $\mathcal{F}_W$ is linear in rW ; $\mathcal{X}$ is the coherent-channel force associated with the unrenormal-

ized projected coefficient $D_0(\beta_0)$; and $\mathcal{I}_\Gamma$ is the full retarded/advanced/ lesser influence-kernel force. Its Langreth form contains both $K_\Gamma^R G^<$ and $K_\Gamma^< G^A$; dropping the latter is not a valid thermal saddle. In the chiral basis the relative dissipative kernel is off-diagonal, so $G_{+-}^<$ and $G_{-+}^<$ cannot be discarded. The Supplemental Material gives the explicit expressions and the no-double-counting rule.

A central point, emphasized to avoid any gauge ambiguity, is that the physical amplitude $r = |b_c|$ is gauge-invariant, whereas the phase of b_c is an internal $U(1)$ slave-boson gauge degree of freedom ($b \rightarrow be^{i\theta}$, $f_\sigma \rightarrow f_\sigma e^{-i\theta}$). Consequently no observable conclusion is drawn from the phase of b_c . All spectra, condition numbers, and scale estimates below use the gauge-invariant amplitude $\tilde{\beta} = |\beta_0|r$ for the symmetric finite-drive diagnostics. If a numerical saddle is represented with a complex b_c , that phase is only a convention for the auxiliary fields and is not used as a physical source of the resonance shift.

C. Lesser Green's Function and Thermal Occupation

The physical lesser Green's function is fixed by the thermal bath. For a bilinear impurity problem coupled to a reservoir at temperature T_b , the Keldysh construction gives

$$G_{\text{bath}}^<(\omega) = G^R(\omega) \Sigma_{\text{bath}}^<(\omega) G^A(\omega) = f(\omega) [G^A(\omega) - G^R(\omega)]. \quad (55)$$

with $f(\omega) = [e^{\omega/T_b} + 1]^{-1}$. This bath-controlled object is the $G^<$ used for the physical occupations, with $n_f = -i \text{Tr} G^<(t, t)$. Equivalently, $\Sigma_{\text{bath}}^< = i f \Gamma_{\text{bath}}$ and $i(G^R - G^A) = G^R \Gamma_{\text{bath}} G^A$. This sandwich form retains the complete channel matrix and introduces no separate biorthogonal normalization factor.

For comparison only, one can also assign Fermi occupations to the biorthogonal eigenmodes of $\tilde{H}(k)$,

$$G_{\text{eig}}^<(k) = iR(k) \text{diag}[f(\text{Re } \mathcal{E}_n)] L^\dagger(k), \quad (56)$$

where $L^\dagger \tilde{H} R = \text{diag}(\mathcal{E}_n)$ and $L^\dagger R = \mathbb{I}$. This eigenmode construction is not a physical thermal distribution function for the bath-coupled impurity; it is used only as an auxiliary diagnostic of nonorthogonality and must be interpreted with care near an EP.

D. Shifted Friedel-Type Resonance Condition

In the standard Hermitian slave-boson mean field at particle-hole symmetry ($\epsilon_\xi = -U/2$), the Kondo resonance is pinned to the Fermi level, $\omega_{\text{res}} = 0$, because the two spin occupations are equal and the saddle point fixes $\tilde{\epsilon}_\xi = 0$. In the present driven problem the

coarse-grained self-energy produces spin-selective imaginary shifts $x_\sigma = \zeta_\sigma \Gamma_{\text{PT}}$ and therefore different effective widths in the two spin channels. The physical resonance position is then most conservatively described by the gauge-invariant condition

$$\omega_{\text{res}} \simeq \text{Re}(E_{\text{EP}}) \simeq \text{Re}(\tilde{\epsilon}_\xi), \quad (57)$$

valid when the low-energy line shape remains approximately Lorentzian. This is a shifted resonance condition rather than a claim that the gauge-dependent phase of b_c is observable. The shift is controlled by the renormalized level $\tilde{\epsilon}_\xi$, the spin-dependent widths, and the self-consistent amplitude $\tilde{\beta} = |\beta_0| |b_c|$. In the strongly non-Lorentzian regime the observed peak need not coincide exactly with this reference, and we therefore treat the deviation as a line-shape effect rather than as a violation of a sum rule. In the frozen kernel used for the plotted spectra, the local-moment reference has $\text{Re}(\tilde{\epsilon}_\xi) \simeq -1.0$ for $\epsilon_\xi = -U/2 = -1.0$; a fully self-consistent finite-temperature SBMF closure, when a finite- r branch exists, would replace this by its computed saddle-point value.

E. Near-EP Analytic Screening-Scale Estimate

Near an isolated EP, the individual spectral projectors grow as the left and right eigenvectors coalesce, while the complete resolvent remains the finite Jordan expression derived in Sec. IX. We use the condition number κ_{imp} only as a diagnostic of nonnormality and EP distance. The screening estimate is controlled instead by the eigenvalues of the complete channel-projected Schrieffer–Wolff matrix. We write

$$T_K^{\text{est}} = D_{\text{uv}} \exp \left[-\frac{1}{\chi_K j_{\text{scr}}} \right], \quad (58)$$

$$j_{\text{scr}} = \max \{ \text{Re } j_\alpha : j_\alpha \in \text{spec}(\rho_b J^{\text{bio}}), \text{Re } j_\alpha > 0 \}. \quad (59)$$

Here j_α are the eigenvalues of the dimensionless biorthogonal exchange matrix $\rho_b J^{\text{bio}}$, and only positive-real-part eigenvalues are retained as antiferromagnetic screening channels. The coefficient $\chi_K > 0$ fixes the conventional scaling normalization; all plotted estimates use $\chi_K = 1$. A scale increase is possible only if the complete EP-active channel calculation increases such a positive screening eigenvalue. No individual projector residue is inserted. By contrast, a hybridization induced EP generated by a negative scalar product $V^R V^L < 0$ can produce strong spectral nonnormality and a frozen non-Hermitian crossover scale, but it should not automatically be identified with an enhanced thermodynamic Kondo temperature. Supplemental Material gives the corresponding frozen-model scale estimate and explains why the spin-selective sign $x_\uparrow = -x_\downarrow$ alone does not split the scale unless the channel widths, real detunings, or exchange eigenvalues are unequal.

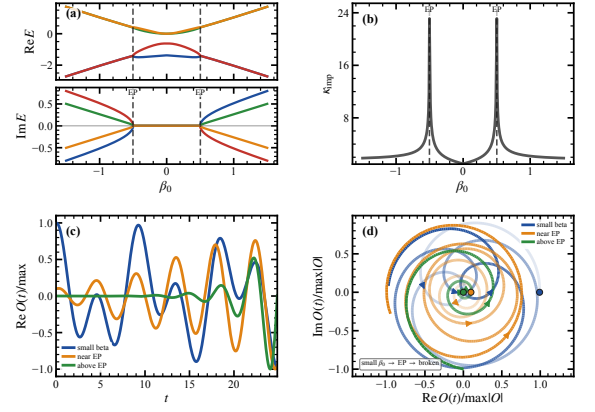


FIG. 1. **Full 4×4 non-Hermitian impurity kernel: spectrum, condition number, and real-time dynamics.** All panels use the gauge-fixed frozen-kernel convention $\tilde{\beta} = |\beta_0| r$ with $\epsilon_\xi = -U/2 = -1.0$, $U = 2.0$, and $T_b = 0.1$. The plotted eigenvalues and eigenvectors are computed from the full rotated 4×4 kernel of Eq. (42), not from the isolated impurity block. **(a)** Real and imaginary parts of the four eigenvalues of $\tilde{H}(k=0)$ versus β_0 . The impurity-related double-root condition at this SOC-degenerate point is exactly the local-core value $\beta_{0,\text{core}}^{\text{EP}} = 0.50$; the discrete condition-number grid peaks at the adjacent point $\beta_0 = 0.5025$. The separately tracked finite- k operational minimum is $\beta_{0,\text{proj}}^{\text{min}} = 0.4975$. **(b)** Condition number $\kappa_{\text{imp}} = \|R\| \|R^{-1}\|$, using the same eigenvector normalization throughout the scan. The finite- $|\beta_0|$ peaks locate eigenvector coalescence; the origin is a normal degeneracy and is not interpreted as an EP. **(c)** Survival-amplitude dynamics for representative points below, at, and above the EP, plotted on a common solver time window. **(d)** Corresponding phase portraits. The change from oscillatory motion to critical coalescence and then to broken-regime overdamped response is the time-domain signature of the 4×4 finite-drive EP.

VI. SPECTRAL PROPERTIES AND PHYSICAL REGIMES

Unless stated otherwise, the final numerical figures use the uniform local-moment convention

$$\epsilon_\xi = -U/2 = -1.0, \quad U = 2.0, \quad T_b = 0.1, \quad (60)$$

with the gauge-fixed projected scale $\tilde{\beta} = |\beta_0| r$. The exact $k=0$ core/dressed value is $\beta_{0,\text{core}}^{\text{EP}} = 0.50$, the separately tracked finite- k operational minimum is $\beta_{0,\text{proj}}^{\text{min}} = 0.4975$, and the discrete condition-number grid peaks at $\beta_0 = 0.5025$. Figure 1 displays the $k=0$ diagnostic near $|\beta_0| \simeq 0.50$.

Figures 1 and 2 display the dynamical fingerprints of the EP-active projected impurity block and of the relative gain-loss-only diagnostic limit, respectively. In the frozen particle-hole-symmetric diagnostic kernel ($\epsilon_\xi = -1.0 = -U/2$, $U = 2.0$, $T_b = 0.1$), the representative coefficients place a symmetric local EP pair at $|\beta_0| \simeq 0.5$, accompanied by a condition-number peak and by a crossover from oscillatory to overdamped local dy-

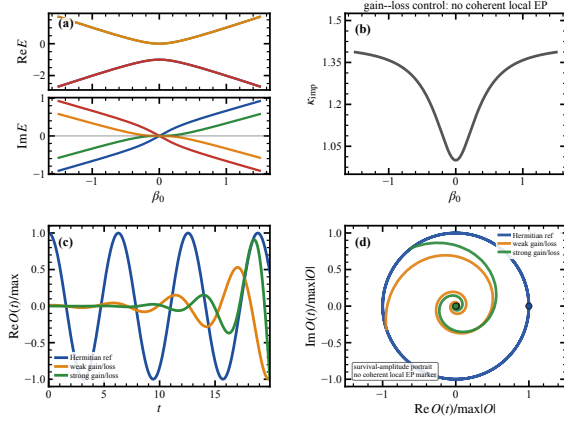


FIG. 2. **Full 4×4 relative gain-loss-only control.** The same impurity-level convention is used as in Fig. 1, $\epsilon_\xi = -U/2 = -1.0$ with $U = 2.0$ and $T_b = 0.1$, but the coherent local splitting/flip channel is switched off. This isolates passive relative gain-loss and bath dressing from the coherent impurity-EP mechanism. (a) Real and imaginary parts of the four eigenvalues of the frozen 4×4 kernel. (b) Condition-number diagnostic. Any residual structure is a passive non-Hermitian bath-dressing diagnostic and is not interpreted as a coherent finite-drive local impurity EP. (c) Survival-amplitude dynamics for the Hermitian reference, weak relative gain-loss, and strong relative gain-loss cases, plotted on a common time window. (d) Phase portraits for the same three cases. These trajectories are shown in a normalized or co-decaying representation; they should not be read as conservation of the Euclidean norm of the isolated relative gain-loss block.

namics. Removing the coherent splitting channel ($\gamma = 0$, Fig. 2) removes the coherent local impurity EP mechanism. The remaining passive relative gain-loss dynamics is retained only as a diagnostic contrast: it should not be interpreted as a separate positive-screening Kondo scale enhancement.

a. Condition-number convention. All condition numbers shown below are computed from the Euclidean right-eigenvector matrix of the full 4×4 channel-projected kernel at the plotted k slice, using a single normalization convention for all values of β_0 . We use κ_{imp} to locate finite- $|\beta_0|$ eigenvector coalescence and to track the distance to the EP. Its absolute magnitude is not used as a standalone physical observable or as an independent definition of the Kondo scale; the physical diagnostics are the spectral line shapes, low-energy scale estimates, occupied fluctuation-dissipation-relation (FDR)-weighted spectra, and frozen channel-projected amplitudes extracted with the same gauge-invariant convention $\tilde{\beta} = |\beta_0||b_c|$.

A. Causal Pre-EP Amplification

The complete causal channel density and the channel-resolved weak-coupling flow provide supporting tests of

the analytic construction; the full panels and adverse controls are shown in Supplemental Figs. S4 and S6. For $\beta_0 = 0.50$, the active/control ratios are 1.0567 at $g_* = 0.30$, 1.0179 at $g_* = 0.40$, 0.9893 at $g_* = 0.50$, and 0.9166 at $g_* = 1$. Thus the larger leading channel-density eigenvalue and early RG trajectory are bounded pre-EP effects in the stated window, but they do not support the stronger statement of an enhanced asymptotic Kondo temperature. The matrix density is positive in the retained passive window, and the result is stable under RG-step refinement. A rate-matched off-EP control can exceed the EP in the largest proper-delay peak, so the Smith delay is not counted as independent evidence of enhancement; this control is shown in the Supplemental Material.

VII. FLUCTUATION-DISSIPATION RELATION AND SPECTRAL FUNCTIONS

For a non-Hermitian system, two inequivalent constructions of $G^<$ arise and it is essential to distinguish them. Supplemental Fig. S7 separates these objects explicitly. The impurity and bath FDR ratios constructed from the reservoir self-energy follow the same thermal Fermi function on spectral support. The eigenmode-based construction is displayed in a separate diagnostic panel because it is not a bath-imposed distribution function; deviations of this diagnostic are therefore not a physical violation of the fluctuation-dissipation relation.

a. Bath-based construction (physical). The physical lesser Green's function is fixed by the thermal bath self-energy. For a bilinear impurity-bath coupling,

$$G_{\text{phys}}^<(\omega) = G^R(\omega) \Sigma_{\text{bath}}^<(\omega) G^A(\omega). \quad (61)$$

If the bath is thermal, then

$$\begin{aligned} \Sigma_{\text{bath}}^<(\omega) &= if(\omega)\Gamma_{\text{bath}}(\omega) \\ &= f(\omega) [\Sigma_{\text{bath}}^A(\omega) - \Sigma_{\text{bath}}^R(\omega)]. \end{aligned} \quad (62)$$

Combining this with the Dyson identity for the bath-induced broadening gives

$$G_{\text{phys}}^<(\omega) = f(\omega) [G^A(\omega) - G^R(\omega)]. \quad (63)$$

Thus the physical impurity FDR is inherited from the reservoir and is the $G^<$ used in the SBMF self-consistency. Charge conservation is checked through the steady-state collision integral

$$\mathcal{I}_N = \int \frac{d\omega}{2\pi} \text{Tr}[\Sigma^<G^> - \Sigma^>G^<] = 0, \quad (64)$$

which vanishes for the same complete thermal bath self-energy. This is a nontrivial number-continuity check; diagonalizing the non-Hermitian kernel and assigning occupations mode by mode need not satisfy it.

b. Eigenmode-based construction (comparison). $G_{\text{eig}}^< = iR \text{diag}(f(\text{Re } \mathcal{E}_n)) L^\dagger$ assigns Fermi occupations $f(\text{Re } \mathcal{E}_n)$ to the non-Hermitian eigenvalues $\mathcal{E}_n$. Because the eigenvalues are complex, this construction does not correspond to the thermal occupations imposed by the bath. Consequently $G_{\text{eig}}^<$ is not interpreted as a physical distribution function. It is retained only as an auxiliary diagnostic of the nonorthogonal eigenbasis and is not used in the SBMF self-consistency or in the physical FDR analysis.

c. Impurity spectral function. The impurity spectral function is evaluated from the retarded and advanced Green's functions as

$$A_{\text{imp}}(\omega) = -\frac{1}{2\pi} \text{Im Tr}[G^R(\omega) - G^A(\omega)]_{\text{imp}}. \quad (65)$$

We use this bath-consistent construction throughout the spectral analysis. In contrast, using only $-\text{Im Tr } G^R/\pi$ can introduce spurious features from relative-kernel poles in a non-Hermitian representation. The spectral results below are therefore computed with the same retarded, advanced, and bath-controlled lesser functions used in the self-consistency.

d. Gauge convention for spectral functions. The physical self-energy from the coarse-graining is first decomposed into a common bath shift/damping, the passive $\mathcal{PT}$ core of Eq. (31), and the Kramers–Kronig detuning of Eq. (30). The off-shell auxiliary sector supplies the real spin-odd shift, while the energy-dependent Dirac bath converts this shift into the relative spin-dependent width Γ_{PT} . The effective Hamiltonian is therefore written in terms of the real, gauge-invariant amplitude $r = |b_c|$ and the projected channels $\Gamma_{\text{PT}} = r^2 \Gamma_{\text{PT}}^{(0)}(\beta_0)$, $\Delta_{\text{eff}} = r^2[\Delta_0^{(0)} + c_\Delta^{(0)}\beta_0]$, and $V_{\text{eff}} = rW$. The phase of the auxiliary slave-boson field is not used separately in spectral calculations. With this convention, the same H_{ss} is used for eigenvalues, condition numbers, and spectral functions, and no observable is attributed to the gauge-dependent phase of b_c .

VIII. REAL-TIME DYNAMICS

The real-time diagnostic is the complex survival amplitude

$$O(t) = \langle d_\uparrow | e^{-iHt} | d_\uparrow \rangle, \quad (66)$$

evaluated for the same gauge-fixed frozen kernel used in the spectral plots. We show both $\text{Re } O(t)$ and the phase portrait $[\text{Re } O(t), \text{Im } O(t)]$. This is a non-unitary propagator diagnostic of the local EP; it is not used as a normalized thermal expectation value. For Fig. 1(c),(d) we use $\epsilon_\xi = -U/2 = -1.0$, $U = 2.0$, $T_b = 0.1$, and $\beta_{0,\text{core}}^{\text{EP}} = 0.50$. The evolution shows a direct dynamical signature of the EP:

- *Below EP* ($\beta_0 < \beta_{0,\text{core}}^{\text{EP}}$, local splitting real): eigenvalues are non-degenerate; the evolution shows underdamped oscillations with a frequency set by

$\text{Re}(E_+ - E_-) = 2\text{Re } s$ and a decay rate set by $\text{Im}(E_\pm)$.

- *Above EP* ($\beta_0 > \beta_{0,\text{core}}^{\text{EP}}$, local splitting imaginary): eigenvalues have opposite imaginary parts; the gain mode dominates and the evolution shows overdamped decay.
- *Without flip* ($\gamma = 0$, Fig. 2): the isolated relative gain-loss block has eigenvalues $\tilde{\epsilon}_\xi \pm i\Gamma_{\text{PT}}$ and is therefore in the broken sector for $\Gamma_{\text{PT}} \neq 0$. The plotted bounded trajectories are normalized or co-decaying diagnostics of the full bath-coupled kernel, not evidence for conservation of the Euclidean norm.

The contrast between the normalized no-flip diagnostic and the spiraling dynamics with coherent splitting demonstrates that both projected impurity channels of $\tilde{H}$ —the coherent splitting Δ_{eff} and the relative decay channel $i\Gamma_{\text{PT}}$ —are necessary for the full finite-drive EP structure. In the channel-projected parametrization, these channels are measured relative to the common scale $\tilde{\beta}$ but should be viewed as distinct projected self-energy components. Equality of their coefficients is a symmetric representative limit, not a symmetry requirement.

IX. CAUSAL SCATTERING AND LINEARIZED CONTACT ALGEBRA

A. Complete scattering and Jordan response

We use the standard Fisher–Lee and Wigner–Smith constructions only as causal scattering diagnostics of the projected kernel [39–41]; they are not introduced here as new formalisms. The basis labels must be kept explicit. In the original spin basis $\Psi_s = (d_\uparrow, d_\downarrow)^T$, the compensated Markov kernel and its positive rate matrix are

$$\begin{aligned} h_s &= \Delta_{\text{eff}}\sigma_x + i\Gamma_{\text{PT}}\sigma_z - i\bar{\Gamma}I, \\ \Gamma_s &= i(\Sigma_s^R - \Sigma_s^A) = 2(\bar{\Gamma}I - \Gamma_{\text{PT}}\sigma_z) \succeq 0. \end{aligned} \quad (67)$$

The Hadamard matrix U_H that defines the fixed chiral fields $\Psi_\chi = U_H^\dagger \Psi_s$ gives

$$\begin{aligned} h_\chi &= U_H^\dagger h_s U_H \\ &= \Delta_{\text{eff}}\sigma_z + i\Gamma_{\text{PT}}\sigma_x - i\bar{\Gamma}I, \\ \Gamma_\chi &= U_H^\dagger \Gamma_s U_H \\ &= 2(\bar{\Gamma}I - \Gamma_{\text{PT}}\sigma_x). \end{aligned} \quad (68)$$

For either representation $\nu = s, \chi$,

$$S_\nu(\omega) = I - i\Gamma_\nu^{1/2} G_\nu^R(\omega) \Gamma_\nu^{1/2}, \quad S_\chi = U_H^\dagger S_s U_H, \quad (69)$$

and, in the passive window, S_ν is unitary. The Wigner–Smith matrix

$$Q_\nu(\omega) = -iS_\nu^\dagger(\omega) \partial_\omega S_\nu(\omega) \quad (70)$$

is Hermitian and transforms covariantly. Thus its trace and eigenvalues, like the eigenphases of S , are basis invariant. Individual entries such as S_{+-} are representation dependent and are not used as evidence for enhancement. The basis-resolved Green functions and coefficients are worked out in the Supplemental Material.

For the compensated unbroken Markov core, $s = \sqrt{\Delta_{\text{eff}}^2 - \Gamma_{\text{PT}}^2} \in \mathbb{R}$ and the two causal scattering phases give

$$\begin{aligned} \text{Tr } Q(\tilde{\epsilon}_\xi) &= \frac{4\bar{\Gamma}}{s^2 + \bar{\Gamma}^2}, \\ \frac{\text{Tr } Q_{\Gamma_{\text{PT}}}(\tilde{\epsilon}_\xi)}{\text{Tr } Q_0(\tilde{\epsilon}_\xi)} &= \frac{\Delta_{\text{eff}}^2 + \bar{\Gamma}^2}{\Delta_{\text{eff}}^2 - \Gamma_{\text{PT}}^2 + \bar{\Gamma}^2} > 1. \end{aligned} \quad (71)$$

Thus, at fixed Δ_{eff} and $\bar{\Gamma}$, the invariant scattering phase density is analytically concentrated at the resonance center as the EP is approached. Its integrated weight remains fixed, and linewidth- or rate-matched adverse controls need not share the peak ordering. We call Eq. (71) path-specific causal preamplification, not an enhanced thermodynamic Kondo temperature.

The cancellation responsible for its regularity is transparent for the traceless core $h = \Delta_{\text{eff}}\sigma_x + i\Gamma_{\text{PT}}\sigma_z$, $h^2 = s^2I$. Although the individual biorthogonal projectors $P_\pm = (I \pm h/s)/2$ grow as $1/s$, their complete resolvent is

$$\frac{P_+}{z-s} + \frac{P_-}{z+s} = \frac{zI+h}{z^2-s^2}. \quad (72)$$

At the EP, $h_{\text{EP}}^2 = 0$, and

$$(zI - h_{\text{EP}})^{-1} = \frac{I}{z} + \frac{h_{\text{EP}}}{z^2}, \quad (73)$$

so the coalescence produces a finite simple-plus-double-pole Jordan response. An independent finite- U biorthogonal NRG audit resolves the same matrix direction in branch-tracked transition residues: the channel trace is strongly suppressed while the channel-odd and symmetric off-diagonal components retain the nilpotent Jordan structure. This is a matrix-residue statement, not a universal interaction-induced pole-splitting law. For the compensated Markovian kernel,

$$\begin{aligned} \text{Tr } Q_{\text{EP}}(\omega) &= \frac{4\bar{\Gamma}}{(\omega - \text{Re } E_{\text{EP}})^2 + \bar{\Gamma}^2}, \\ \int \frac{d\omega}{2\pi} \text{Tr } Q_{\text{EP}} &= 2. \end{aligned} \quad (74)$$

The numerical implementation satisfies scattering unitarity to 1.33×10^{-15} and the trace-delay/determinant identity to 4.97×10^{-14} . A rate-matched off-EP control can nevertheless have a larger maximum proper delay, so this observable diagnoses causal pole coalescence but is not, by itself, evidence of many-body enhancement.

The same cancellation occurs in the finite- U virtual-charge sum. At particle-hole symmetry, with $A = U/2$

and a channel-independent bare vertex W ,

$$\begin{aligned} J_{\text{SW}}(s) &= 2W^2 [(AI - h)^{-1} + (AI + h)^{-1}] \\ &= \frac{4AW^2}{A^2 - s^2}I, \quad J_{\text{SW}}(0) = \frac{8W^2}{U}. \end{aligned} \quad (75)$$

Thus neither a Green function nor an exchange vertex acquires an independent Petermann or condition-number multiplier. This causal scattering statement does not assume integrability. We first record its exact one-particle phase condition and then isolate the additional low-energy assumptions under which the finite- U contact problem has a genuine multiparticle Yang-Baxter algebra.

B. One-particle contact-phase quantization

Let $\mathbf{s}_a(\omega)$, $a = 1, 2$, be the two eigenvalues of the complete unitary matrix $S(\omega)$ in Eq. (69). In the passive window

$$\begin{aligned} S(\omega)u_a(\omega) &= \mathbf{s}_a(\omega)u_a(\omega), \\ \mathbf{s}_a(\omega) &= e^{2i\delta_a(\omega)}, \quad |\mathbf{s}_a| = 1. \end{aligned} \quad (76)$$

The index a labels scattering eigenphases, not necessarily the fixed chiral channels: although the bare vertex is diagonal after the Hadamard rotation, the internal $i\Gamma_{\text{PT}}$ term generally produces off-diagonal channel conversion. After linearizing the radial propagation in the stated finite scattering window, a ring of length $\mathcal{L}$ has the exact one-particle finite-size condition

$$\det[I - e^{ik\mathcal{L}}S(k)] = 0 \iff e^{ik_{ja}\mathcal{L}}\mathbf{s}_a(k_{ja}) = 1. \quad (77)$$

This is an exact one-particle scattering statement. It is not, by itself, a multiparticle Bethe ansatz or a Yang-Baxter theorem.

The calculation-level consistency check is exact for the full matrix:

$$\begin{aligned} K_{\text{ph}}(\omega) &= \sum_{a=1}^2 \frac{1}{2\pi i} \partial_\omega \ln \mathbf{s}_a(\omega) = \frac{1}{2\pi i} \partial_\omega \ln \det S(\omega) \\ &= \frac{1}{2\pi} \text{Tr } Q(\omega) \equiv \Delta\rho_{\text{scatt}}(\omega). \end{aligned} \quad (78)$$

The numerical gate reported in Supplemental Fig. S6 gives $\int d\omega K_{\text{ph}} = 1.99236$ over the displayed finite integration window, while the complete analytic pole count is 2. This identity directly validates the phase quantization against the causal scattering calculation; it does not define a thermodynamic Kondo temperature. The same full-matrix gate finds the two scattering eigenvalues unimodular to 1.11×10^{-15} , and an independent finite-difference evaluation of the phase identity agrees to 5.53×10^{-9} . The earlier value 0.9213 is retained in the Supplemental coefficient audit only as a spin-representation matrix element (equivalently, at the resonance center, the magnitude of the imaginary part of the two scattering eigenvalues); it is not called a fixed-chiral-channel conversion probability.

C. Controlled branchwise linearization and CPT similarity

The two SOC branches in Eq. (42) are shifted copies of the same parabola,

$$\varepsilon_{c,\eta}(k) = \alpha_b \left(k + \eta \frac{\lambda}{2\alpha_b} \right)^2 - \frac{\lambda^2}{4\alpha_b}, \quad \eta = \pm. \quad (79)$$

Linearizing each branch about a counterpropagating Fermi point gives

$$\varepsilon_{c,\eta}(k_{\eta F} + q) = \mu + \eta v q + O(q^2). \quad (80)$$

After subtracting μ and folding the left mover, both components have the same right-moving kinetic term. The controlled reduction used from this point onward assumes equal folded velocities, drops $O(q^2)$ curvature, freezes the component matrix, and retains the momentum-independent scalar hybridization obtained in Eq. (41). It is

$$\begin{aligned} H_{\text{lin}}(M) = & -iv \int_0^{\mathcal{L}} dx \bar{\chi}_a \partial_x \chi_a \\ & + \bar{d}_a (\epsilon_d \delta_{ab} + M_{ab}) d_b + U n_{d+} n_{d-} \\ & + V [\bar{\chi}_a(0) d_a + \bar{d}_a \chi_a(0)]. \end{aligned} \quad (81)$$

$$M = D\sigma_z + iG\sigma_x, \quad D \equiv \Delta_{\text{eff}}, \quad G \equiv \Gamma_{\text{PT}}. \quad (82)$$

Here a bar denotes the canonical dual; under a non-unitary change of components it transforms with the inverse matrix, not the Hermitian adjoint.

Away from the EP, $s^2 = D^2 - G^2 \neq 0$, the constant matrix is diagonalized by

$$S_C = e^{\theta \sigma_y / 2}, \quad \tanh \theta = \frac{G}{D}, \quad S_C^{-1} M S_C = s \sigma_z. \quad (83)$$

For $|G| < |D|$, θ is real and S_C is the usual non-unitary CPT/metric similarity; outside that region Eq. (83) is its complex $GL(2, \mathbb{C})$ continuation. The simultaneous canonical transformation

$$\chi = S_C \tilde{\chi}, \quad d = S_C \tilde{d}, \quad \bar{\chi} = \tilde{\chi} S_C^{-1}, \quad \bar{d} = \tilde{d} S_C^{-1} \quad (84)$$

leaves the kinetic term and scalar hybridization unchanged. It also leaves the interaction unchanged because

$$n_{d+} n_{d-} = \frac{1}{2} N_d (N_d - 1), \quad N_d = \bar{d}_a d_a, \quad (85)$$

and N_d is invariant under Eq. (84). Thus finite U is compatible with, rather than an obstruction to, this component rotation. At $s = 0$, $M \neq 0$ is nilpotent and the diagonalizer is singular; the EP must be treated in the Jordan or boundary-twist form below, not as a diagonal CPT frame.

D. Finite- U two-electron contact algebra

Let $\mathcal{A}_{12}$ and $\mathcal{A}_{21}$ be the two-component pseudospin amplitudes in the two coordinate-ordering sectors. Eliminating the empty, singly occupied, and doubly occupied impurity contact amplitudes gives the triplet and singlet matching conditions

$$\begin{aligned} \mathcal{A}_{21}^{(t)} &= \mathcal{A}_{12}^{(t)}, \\ [\Delta \mathcal{B} + i 2U \Gamma_A] \mathcal{A}_{21}^{(s)} &= [\Delta \mathcal{B} - i 2U \Gamma_A] \mathcal{A}_{12}^{(s)}, \\ \Delta \mathcal{B} &= \mathcal{B}(p) - \mathcal{B}(q), \end{aligned} \quad (86)$$

$$\mathcal{B}(p) = p(p - 2\epsilon_d - U), \quad \Gamma_A = \frac{V^2}{2v}.$$

The explicit elimination is given in the Supplemental Material. With $\Pi_t = (I + P_{12})/2$ and $\Pi_s = (I - P_{12})/2$, these two equations combine to the exact finite- U Anderson contact matrix [42–45],

$$\begin{aligned} R_{12}(p, q) &= \Pi_t + \frac{\Delta \mathcal{B} - i 2U \Gamma_A \Pi_s}{\Delta \mathcal{B} + i 2U \Gamma_A} \\ &= \frac{\Delta \mathcal{B} I_{12} + i 2U \Gamma_A P_{12}}{\Delta \mathcal{B} + i 2U \Gamma_A}. \end{aligned} \quad (87)$$

For $U \neq 0$, define the dressed Anderson rapidity

$$u(p) = \frac{\mathcal{B}(p)}{2U \Gamma_A}, \quad x = u(p) - u(q). \quad (88)$$

Up to the displayed scalar normalization,

$$R_{12}(x) = \frac{x I_{12} + i P_{12}}{x + i} = \frac{1}{x + i} \begin{pmatrix} x + i & 0 & 0 & 0 \\ 0 & x & i & 0 \\ 0 & i & x & 0 \\ 0 & 0 & 0 & x + i \end{pmatrix}, \quad (89)$$

in the ordered basis $\{|++\rangle, |+-\rangle, |-+\rangle, |--\rangle\}$. The identity

$$\mathcal{B}(p) - \mathcal{B}(q) = (p - q)(p + q - 2\epsilon_d - U) \quad (90)$$

shows why the bare pair-energy factor does not obstruct factorization: the exact difference variable is $u(p) - u(q)$. The map $p \mapsto u(p)$ is two-to-one and its $U \rightarrow 0$ limit is singular; neither fact affects the finite- U matrix identity, while the quadratic benchmark must be handled separately.

Permutation algebra gives the ordinary Yang–Baxter equation

$$\begin{aligned} R_{12}(u - v) R_{13}(u - w) R_{23}(v - w) \\ = R_{23}(v - w) R_{13}(u - w) R_{12}(u - v). \end{aligned} \quad (91)$$

Moreover, for every $S \in GL(2, \mathbb{C})$,

$$(S \otimes S)^{-1} R_{12}(x) (S \otimes S) = R_{12}(x), \quad (92)$$

because $[P_{12}, S \otimes S] = 0$. In particular, the CPT/metric transformation S_C preserves the ordinary YBE. It does not generate a new bulk R -matrix; the non-Hermitian datum resides in the boundary twist.

E. Boundary twist and precise scope

Let $H_A = H_{\text{lin}}(0)$. Its scalar kinetic, hybridization, and interaction terms commute with the complexified global pseudospin charge

$$Q_M = \int_0^{\mathcal{L}} dx \bar{\chi} M \chi + \bar{d} M d. \quad (93)$$

To display the transverse terms explicitly, define

$$\mathcal{J}^\mu = \frac{1}{2} \int_0^{\mathcal{L}} dx \bar{\chi} \sigma_\mu \chi \frac{1}{2} \bar{d} \sigma_\mu d, \quad \mathcal{J}^\pm = \mathcal{J}^x \pm i \mathcal{J}^y. \quad (94)$$

For $M = D\sigma_z + iG\sigma_x$,

$$Q_M = 2D\mathcal{J}^z + 2iG\mathcal{J}^x = 2D\mathcal{J}^z + iG(\mathcal{J}^+ + \mathcal{J}^-). \quad (95)$$

Thus the nonzero ladder components, often denoted $S^\pm$, are retained as parts of a conserved *global* charge. Away from the EP the canonical $GL(2, \mathbb{C})$ transformation Eq. (84) reduces this charge to $2s\mathcal{J}^z$ without changing the rational bulk R -matrix. At the EP the diagonalizer is not used: Q_M is conjugate to a nonzero nilpotent ladder generator and the boundary twist remains unipotent and invertible.

On a finite ring, the bounded multiplication operator $e^{-iMx/v}$ maps $H_A + Q_M$ to the impurity form Eq. (81) with boundary twist

$$\mathcal{G} = e^{iM\mathcal{L}/v}. \quad (96)$$

At the EP, $M^2 = 0$, so $\mathcal{G} = I + iM\mathcal{L}/v$ is unipotent but remains invertible. Taking $L_{aj}(z) = R_{aj}[z - u(p_j)]$, inserting $\mathcal{G}$ in the auxiliary monodromy, and using Eq. (92) gives RLL, twisted RTT, and $[t_{\mathcal{G}}(z), t_{\mathcal{G}}(w)] = 0$ for arbitrary particle number. The bounded-domain proof and the many-body Jordan consequences are the subject of Ref. [36]; they are not inferred here from a singular diagonalizer. In particular, that work proves explicitly that the transverse ladder terms generate, rather than lift, the Jordan chain on the integrable submanifold. It also solves the twisted nested Bethe equations and locates the universal descendant-root contraction at the projective point $\lambda = \infty$, with scaled coordinates fixed by a generalized Laguerre polynomial. It does not claim coalescence of the finite Anderson charge roots.

This exact bridge requires equal folded velocities, a linear wide band, constant M , scalar momentum-independent hybridization, and the standard component-symmetric local U . The complete driven model also contains curvature, energy dependence, causal detuning, and potentially unequal velocities or non-scalar projected vertices. Those terms lie outside the proof and can shift or split the symmetry-protected EP. Thus the result establishes a controlled integrable low-energy submanifold, not the Yang–Baxter integrability of the full driven nonequilibrium kernel.

F. Frozen Schrieffer–Wolff screening estimate

The Yang–Baxter identity is algebraic and does not itself determine a thermodynamic screening scale. For comparison with the numerical weak-flow diagnostic, we retain only the conventional frozen Schrieffer–Wolff and poor-man’s-scaling estimate. In a positive screening channel,

$$\mathcal{C}_a \equiv |\tilde{\epsilon}_{d,a}|^{-1} + |\tilde{\epsilon}_{d,a} + U|^{-1}, \quad J_a^{\text{SW}} = 2|W_a|^2 \mathcal{C}_a. \quad (97)$$

Writing $\tilde{\Gamma}_a = \pi \rho_{b,a}(0)|W_a|^2$, the corresponding estimate is

$$T_{K,a}^{\text{SW(est)}} \simeq D_{\text{uv},a} \exp \left[-\frac{\pi}{2\chi_K \tilde{\Gamma}_a \mathcal{C}_a} \right]. \quad (98)$$

At particle–hole symmetry,

$$T_{K,a}^{\text{SW(est)}} \simeq D_{\text{uv},a} \exp \left[-\frac{\pi U}{8\chi_K \tilde{\Gamma}_a} \right]. \quad (99)$$

SOC enters through the channel density, velocity Jacobian, principal-value level shift, and overlap,

$$\tilde{\Gamma}_a(\lambda) = \pi \rho_{b,a}(0; \lambda) |W_a|^2, \quad \rho_{b,a}(0; \lambda) = \sum_{k_*} \frac{k_*/(2\pi)}{|\partial_k \varepsilon_a(k; \lambda)|_{k=k_*}}. \quad (100)$$

Thus the analytic estimate can increase relative to a Hermitian reference only when the EP-active projection increases a *positive* channel width or exchange eigenvalue, or moves an ordinary charge denominator within the local-moment window. It does not acquire an independent $1/s$ factor. For a matrix exchange the corresponding statement is obtained from the largest positive real eigenvalue of $\rho_b J$; negative or complex eigenvalues are not called Kondo screening.

X. SCOPE OF THE NEAR-EP SCALE ESTIMATE

Near the EP, the condition number $\kappa_{\text{imp}} \sim C/|s|$ measures eigenvector nonorthogonality. It is useful for locating the projected-core degeneracy but is not a spectral residue, exchange vertex, or quasiparticle weight. The singular parts cancel in the complete Green function [Eq. (72)] and in the finite- U charge sum [Eq. (75)]. Consequently no Z^{bio} or Petermann multiplier is inserted into a propagator, diagrammatic vertex, or Kondo exponent.

On the exact linearized submanifold, the local U preserves the complexified pseudospin symmetry and therefore does not cut off this algebraic Jordan limit. This does not require a divergent observable: the complete resolvent and charge sum remain finite. In the complete

driven kernel, curvature, velocity mismatch, frequency dependence, or component-dependent vertices can move or split the EP, but their effect must be calculated microscopically rather than represented by a universal interaction floor.

Equation (98) retains the conventional frozen Schrieffer–Wolff scale estimate and permits an EP-assisted enhancement relative to a matched Hermitian reference only when the projected Dirac/SOC channel increases a positive screening width or exchange eigenvalue. That conditional estimate is conceptually distinct both from multiplying a Hermitian scale by a biorthogonal norm and from the Yang–Baxter identity.

The direct numerical evidence is narrower. The leading eigenvalue of the complete causal channel density is larger in the chosen Fermi window, and the largest exchange eigenvalue initially grows faster than in the matched Hermitian control. Supplemental Fig. S6 shows that this ordering reverses before strong coupling. The low-temperature slave-boson saddle in Supplemental Fig. S2 validates the r^2 scaling and charge constraint, but the same closure has no finite- r branch at $T_b = 0.1$. We therefore use “pre-EP amplification” only for the demonstrated spectral/weak-flow response. The estimate in Eq. (98) remains conditional on a positive screening channel, and no enhanced thermodynamic Kondo temperature is claimed.

XI. CONCLUSIONS

We have shown that a strictly Hermitian, periodically driven Dirac impurity develops an emergent passive non-Hermitian retarded kernel by a controlled two-stage mechanism: off-shell angular harmonics produce a real spin-odd shift, and the energy-dependent Dirac bath converts that shift into a relative decay channel. After common damping is removed, the compensated kernel contains the passive $\mathcal{PT}$ core of Eq. (31); its Kramers–Kronig partner supplies the causal detuning floor.

The focused conclusions are fourfold. First, the projected channels scale as r^2 while the time-independent impurity–bath vertex scales as rW . Consequently the local core EP is exactly invariant for $r > 0$, whereas a finite- k dressed minimum must be tracked. The low-temperature saddle confirms this structure and charge continuity, but the same closure has no finite- r branch at $T_b = 0.1$; the main spectra are therefore stated explicitly as frozen-kernel diagnostics.

Second, the complete Green function acquires no independent divergent biorthogonal-projector prefactor at coalescence. The divergent pieces of the individual projectors cancel and form the finite-coefficient Jordan simple-plus-double-pole resolvent, Eq. (73). An operator-resolved finite- U biorthogonal NRG audit independently preserves this matrix structure: the branch-tracked double-pole coefficient is nearly perfectly aligned with the nilpotent EP direction while its trace is suppressed by

more than three orders of magnitude. The same analytic cancellation keeps the particle–hole-symmetric finite- U Schrieffer–Wolff exchange finite. Neither the numerical pole splittings nor the complex quartic diagnostic show a stable universal interaction law. Hence biorthogonal normalization does not multiply a propagator, a diagrammatic vertex, or a screening exponent, and no universal interaction-induced EP floor is inferred.

Third, the positive bath rate matrix gives a unitary Fisher–Lee scattering matrix. Its Wigner–Smith trace develops the analytic EP coalescence profile of Eq. (74), providing a causal pole and linewidth fingerprint. Because a rate-matched off-EP control can have a larger proper delay, this profile is not promoted to independent evidence of many-body enhancement. Its exact one-particle contact-phase condition is tied to the same causal object through $K_{\text{ph}} = \text{Tr } Q/(2\pi)$. The direct small-width channel-resolved RG calculation shows bounded pre-EP amplification at weak thresholds, but the ordering reverses before strong coupling; an enhanced thermodynamic Kondo temperature is not established.

Fourth, branchwise linearization and folding identify a precise integrable submanifold of the physical model. The finite- U contact matching yields the exact rational Anderson matrix Eq. (87) in the dressed rapidity Eq. (88). Its $GL(2, \mathbb{C})$ covariance shows directly that an off-EP CPT/metric similarity preserves YBE. At the EP the diagonalizer is singular, but the boundary twist is invertible and unipotent. The nonzero transverse pseudospin ladders are retained as components of the conserved global charge; the companion work [36] develops their diagonal-twist reduction away from the EP and their unipotent RLL/RTT and many-body Jordan realization at it. Curvature and the energy-dependent driven self-energy remain outside this theorem, so the full microscopic model is not claimed to be Yang–Baxter integrable.

CODE AND DATA AVAILABILITY

The numerical scripts and tabulated data used to generate the main and Supplemental figures are included in the accompanying reproducibility archive. The NRG portion contains the independent propagation of $L^\dagger cR$ and $L^\dagger c^\dagger R$, the no-truncation regression, branch tracking, the comprehensive scaling audit, and the exported data used for Supplemental Fig. S8. The source archive also includes `verify_finiteU_contact_algebra.py`, an independent check of the finite- U contact matrix, CPT covariance, and three-particle YBE. A version-controlled copy of the code, data, figure outputs, and manuscript sources is available in the GitHub repository `VMKPHYSMATH/emergent-pt-dirac-impurity` [46]. Release v1.0.1 is archived at doi:10.5281/zenodo.21434683; the all-versions record is doi:10.5281/zenodo.21434682. Release v1.0.1 predates the present biorthogonal-resolvent and causal-RG corrections and is therefore su-

perseded by the accompanying revision archive.

SUPPLEMENTAL MATERIAL

Detailed derivations of the auxiliary-mode influence functional, Keldysh mean-field equations, physical fluctuation–dissipation construction, complete biorthogonal and finite- U charge-resolvent identities, causal matrix-RG validation, the finite- U two-electron contact algebra, and parameter tables are provided in the Sup-

plemental Material [47]. The main text remains self-contained by stating the effective projected kernel, physical spectrum, and claim boundary explicitly.

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[47] (2026), see Supplemental Material at [URL will be inserted by publisher] for detailed Wigner and Schur-complement derivations, Keldysh and screening constructions, low-temperature saddle, causal scattering validation, finite- U contact algebra, the operator-resolved NRG residue audit, and Figs. S1–S8.